

# Teorema de diferenciación de Lebesgue en $\mathcal{L}^1$ para casi todo punto

**Lema 1.** Sea  $f \in \mathcal{L}^1(\mathbb{R})$  y  $\lambda > 0$ . Entonces, el conjunto

$$A_\lambda := \left\{ x \in \mathbb{R} : \limsup_{r \rightarrow 0} \left| \frac{1}{2r} \int_{x-r}^{x+r} f(y) dy - f(x) \right| > \lambda \right\} \text{ tiene medida de Lebesgue cero.}$$

**Demostración:** Sea  $\varepsilon > 0$ . Como  $\mathcal{C}_c(\mathbb{R})$  es denso en  $L^1(\mathbb{R})$  (teo-fn-continua-soporte-compacto-dens)  $\exists g \in \mathcal{C}_c(\mathbb{R}) : \|f - g\|_1 \leq \varepsilon\lambda/8$ . Como  $g$  es continua, por el teo-diferenciacion-lebesgue-continuas/?? sabemos que

$$\forall x \in \mathbb{R} : \lim_{r \rightarrow 0^+} \frac{1}{2r} \int_{x-r}^{x+r} g(y) dy = g(x).$$

Se tiene así que  $\left| \frac{1}{2r} \int_{x-r}^{x+r} f(y) dy - f(x) \right| =$

$$\begin{aligned} &= \left| \frac{1}{2r} \int_{x-r}^{x+r} f(y) dy - \frac{1}{2r} \int_{x-r}^{x+r} g(y) dy + \frac{1}{2r} \int_{x-r}^{x+r} g(y) dy - g(x) + g(x) - f(x) \right| \\ &\leq \frac{1}{2r} \int_{x-r}^{x+r} |f(y) - g(y)| dy + \left| \frac{1}{2r} \int_{x-r}^{x+r} g(y) dy - g(x) \right| + |g(x) - f(x)|. \end{aligned}$$

Como  $\lim_{r \rightarrow 0^+} \frac{1}{2r} \int_{x-r}^{x+r} g(y) dy = g(x)$ , tomando el  $\limsup$  cuando  $r \rightarrow 0^+$ , se tiene que

$$\begin{aligned} \limsup_{r \rightarrow 0} \left| \frac{1}{2r} \int_{x-r}^{x+r} f(y) dy - f(x) \right| &\leq \limsup_{r \rightarrow 0} \frac{1}{2r} \int_{x-r}^{x+r} |f(y) - g(y)| dy + 0 + |g(x) - f(x)| \\ &\leq M(f - g)(x) + |g(x) - f(x)|. \end{aligned} \tag{1}$$

Definimos ahora  $B_\lambda := \{x \in \mathbb{R} : M(f - g)(x) > \frac{\lambda}{2}\}$  y  $C_\lambda := \{x \in \mathbb{R} : |g(x) - f(x)| > \frac{\lambda}{2}\}$ .

Entonces,  $A_\lambda \subset B_\lambda \cup C_\lambda$  porque  $x \in (B_\lambda \cup C_\lambda)^c = B_\lambda^c \cap C_\lambda^c \implies x \in A_\lambda^c$  ya que

$$\begin{cases} x \in B_\lambda^c \implies M(f - g)(x) \leq \lambda/2 \\ x \in C_\lambda^c \implies |g(x) - f(x)| \leq \lambda/2 \end{cases} \implies \limsup_{r \rightarrow 0^+} \left| \frac{1}{2r} \int_{x-r}^{x+r} f(y) dy - f(x) \right| \leq \lambda$$

por la desigualdad (??).

Por tanto, por subaditividad de la medida de Lebesgue,  $m(A_\lambda) \leq m(B_\lambda) + m(C_\lambda)$ . Ahora

bien, por la desigualdad de Hardy-Littlewood,  $m(B_\lambda) \leq \frac{3}{\lambda/2} \|f - g\|_1$ . Entonces,

$$\begin{aligned} m(A_\lambda) &\leq \frac{3}{\lambda/2} \|f - g\|_1 + \int_{C_\lambda} 1 \, dy \\ &\leq \frac{6}{\lambda} \|f - g\|_1 + \frac{2}{\lambda} \int_{C_\lambda} |g(x) - f(x)| \, dx \leq \frac{8}{\lambda} \|f - g\|_1 \leq \varepsilon. \end{aligned}$$

Como  $\varepsilon > 0$  era arbitrario, concluimos que  $m(A_\lambda) = 0$ . ■

## Referenciado en

- `teo-diferenciacion-lebesgue-l1`

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